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Version Info V1.29

Rexroth Mobile Electronics & Systems (BODAS)

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Summary

This page contains the details of *Change request* and *Trouble report* for workflow release 40.sml290

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- UDS
- ISOBUS
- XCP
- Ethernet
- MO

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Compiler information

Compiler Version	Applicability
hightec jdp cd_v4.6.6.1-bosch-1.4	☑ RC5-6/40
hightec ifx cd_v4.6.6.1-bosch-1.4	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40

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Manual

Change request		
Work item ID	**#183924**	**Applicability**
Scenario	Manual Improvement for the multi core API page	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	Multicore related APIs (APIs related to data exchange between cores) are present within SYS chapter page.	
New behavior	A separate chapter page named Multicore API page is created and APIs for data exchange between cores are added as sub-chapters.	
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		
Work item ID		**Applicability**

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Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		

Trouble report		
Work item ID	**#181028**	**Applicability**
Scenario	Misleading documentation about error reaction (DTCs) in ECU Self Monitoring	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	There were misleading Error Reaction data related to ECU Voltage Monitoring (ADC Test Voltage monitoring, VDD3V3 ECU internal voltage monitoring)	
New behavior	Error Reaction data related to ECU Voltage Monitoring are corrected now.	
Work item ID	**#180771**	**Applicability**
Scenario	Missing information of DTC-DFC-Environmental Data for the testers	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	Details related to UDS Services ClearDiagnosticInformation (0x14), ReadDTCInformation (0x19), ControlDTCSetting (0x85) are not available	
New behavior	Key details related to UDS Services ClearDiagnosticInformation (0x14), ReadDTCInformation (0x19), ControlDTCSetting (0x85) are available	

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Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		

Release notes	
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Basic

Change request		
Work item ID	**#186306**	**Applicability**
Scenario	Improved position calculation for BODASdriver for PAL sensor variants with PWM output signals.	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	The position calculation with the current implementation does not have sufficient resolution for some projects.	
New behavior	Position calculation improved by an additional 2 decimal places for the PAL PWM input signals.	
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		
Work item ID		**Applicability**

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Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		

Trouble report		
Work item ID	**#179247**	**Applicability**
Scenario	All power outputs are very sporadically locked by inhibit monitoring, resulting in a 50–100 ms shutdown following the migration to s250. This is a rare occasion.	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	All power outputs are locked by inhibit monitoring which switches off all power outputs for the duration of 50ms- 100msec.	
New behavior	The power outputs are not locked by the inhibit monitoring when the UBAT is proper (around 12V). Additional monitoring mechanism is implemented to ensure that UBAT_MON is properly monitored.	
Work item ID	**#183352**	**Applicability**

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Scenario	ECU Reset occurs when calling bds_outDig_get() and bds_outPwm_get() from an untrusted task.	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	Calling bds_outClcU_get_ffi(), bds_outDig_get() and bds_outPwm_get() APIs from an untrusted task results in an ECU reset.	
New behavior	Calling the new FFI-Functions bds_outClcU_get_ffi(), bds_outDig_get_ffi() and bds_outPwm_get_ffi() APIs from an untrusted task results in a correct function call.	
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		

Release notes	
1	BODASdriver library for PR4 Analog sensor, for more details please refer to the manual.

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Release notes	
2	**Header File Relevance Clarification:** To help users easily identify which header files are required from the delivered BSW build package, the relevance of files under the /includes directory has been clarified. **Current /includes Structure** /includes ├── bds_*.h ├── rbr_*.h ├── Com_Cfg_SymbolicNames.h ├── LinIf_Cfg_SymbolicNames.h └── (other internal/generated headers) **User-Relevant Header Files** • bds_*.h – Primary user-facing API headers (most relevant for customer review) • Com_Cfg_SymbolicNames.h, LinIf_Cfg_SymbolicNames.h - CAN and LIN signal symbolic name header files • rbr_*.h – Relevant only if these interfaces are used in the customer's project All other headers are internal and need not be considered unless explicitly referenced by the user. **Impact for This Software Release (V01.29)** *bds_*.h header files:* Only bds_uds.h contains changes. ➜ These changes require user review. All other modifications in this release do not impact users directly. **Upcoming Change in V01.30** A more organized /includes folder structure will be introduced: /includes ├── bds ├── core ├── gen └── rbr

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Release notes	
	Benefits • Clear separation between user-facing and internal header files • Reduced clutter in /includes • Easier file discovery and long-term maintenance
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UDS

Change request		
Work item ID	**171555**	**Applicability**
Scenario	BSW supports new API to identify active UDS CAN channel - bds_uds_getActiveCanChnl	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior		
New behavior	Currently active UDS CAN channel can be retrieved using this API, which is needed for API bds_uds_registerSvCustInstCfgToNvm	
Work item ID	**171569**	**Applicability**
Scenario	BSW supports new API to update the UDS customer instance configuration to NVM during the programming session request - bds_uds_registerSvCustInstCfgToNvm	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior		
New behavior	This function registers a callback function to save the customer's diagnostic instance details during a programming session request ($10 02). These saved details are then shared with the bootloader through NVM. This API is for OEMs who want to use Bosch CB-ME along with OEM-specific UDS instance settings configured in easyConfig.	

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Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		

Trouble report		
Work item ID	**#135761**	**Applicability**
Scenario	Diagnostic tester requests UDS Service 0x19, Sub-function 0x04 to read DTC Snapshot Records.	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	The BSW included a snapshot entry with DTCSnapshotRecordNumber = 0x00, representing an empty/placeholder snapshot. This caused the following issues: • The response started with 00 00, which according to ISO 14229 can be misinterpreted as "more than 255 DataIdentifiers follow". • Testers other than BODAS-service could decode this incorrectly.	

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New behavior	The BSW no longer sends DTCSnapshotRecordNumber = 0x00 or any empty snapshot record. The response contains only valid records: • Record 0x01 – First occurrence snapshot (max 6 signals) • Record 0x02 – Most recent occurrence snapshot (max 6 signals) No zeros or placeholders appear before Record 0x01.	
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		

Release notes	
1	BSW supports new API to identify active UDS CAN channel - bds_uds_getActiveCanChnl

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Release notes	
2	BSW supports new API to update the UDS customer instance configuration to NVM during the programming session request - bds_uds_registerSvCustInstCfgToNvm
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ISOBUS

Change request		
Work item ID	**#180202**	**Applicability**
Scenario	Receive Callback Not Triggered for Fixed PGNs (ACL, Command Address, Name Management, PGN Request, DM13)	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	For several fixed PGNs configured through bds_isobus_initRxDatabox(), the receive callback function is not triggered when messages are sent from an external tester. This applies to both single-frame and BAM (multi-packet) messages, depending on the PGN. As a result, the Rx databox for these PGNs remains in **INIT** state and the message length is not updated, even though the PGN message was successfully sent and observed on the bus. Affected Fixed PGNs: • **0xEEFF** – ACL • **0xFED8** – Command Address (BAM) • **0x9300** – Name Management • **0xEA00** – PGN Request • **0xDF00** – DM13	

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New behavior	The ISOBUS Rx handler has been updated to correctly process all fixed PGNs registered through bds_isobus_initRxDatabox(). For **each** of the fixed PGNs listed above: • The incoming message (single-frame or BAM, depending on PGN) is correctly identified. • TP reassembly for BAM messages (e.g., PGN 0xFED8) is completed reliably. • The associated Rx databox transitions from **INIT (0x0100)** to **VALID (0x0200)**. • The message length is updated based on the received payload. • The callback function registered by the ASW is **triggered as expected**.	
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		

Trouble report	

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Work item ID	**184669**	**Applicability**
Scenario	Producer timing is varying if any CAN message is received	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	1) One producer is configured in RC4x, in CAN 3 500Kbps 2) From tester send a CAN message 0x123456x periodically in 20ms to RC4x CAN3 3) The producer is configured for 20ms, in the tester it is observed that producer timing is varying and not 20ms.	
New behavior	Producer timing does not vary now based on received messages.	
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		

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Release notes	
1	Producer timing is varying if any CAN message is received
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XCP

Change request		
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		

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New behavior		

Trouble report		
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12/40, RC27-18/40 ☐ RC18-12E/40, RC27-18E/40 ☐ RC5-6/40
Old behavior		

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New behavior		

Release notes	
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Ethernet

Change request		
Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		

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New behavior		

Trouble report		
Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		

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New behavior		

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MO

Change request		
Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		
New behavior		
Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		

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New behavior		

Trouble report		
Work item ID	**#178021**	**Applicability**
Scenario	bds_out_po_cfgPwrLs API returns *invalid parameter* in newer workflow (ML270), but works in older workflow (ML230)	☑ RC18-12/40, RC27-18/40 ☑ RC18-12E/40, RC27-18E/40 ☑ RC5-6/40
Old behavior	In application, the API bds_out_po_cfgPwrLs is used to configure a proportional high-side output as a power supply for low-side outputs. • In older workflow (ML230): The API executes successfully, and the outputs are supplied and monitored correctly. • In newer workflow (ML270): The same call to bds_out_po_cfgPwrLs consistently returns “invalid parameter” (return value = 1). The power supply configuration does not. Observed Behavior: • API return value: 1 (invalid parameter). • In some cases, “output disabled” (return value 5) was also observed. • Resetting errors with bds_out_po_resetError has no effect.	
New behavior	• User needs to make sure the MoCSOP checks are completed before they try to actuate outputs. This can be performed by polling **bds_mo_getSopStatus()** API and if the MocSOP checks are completed, this API would return TRUE. • After this, when the outputs are actuated, there will be no errors.	

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Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		
New behavior		

Work item ID		**Applicability**
Scenario		☐ RC18-12E/40, RC27-18E/40
Old behavior		
New behavior		

Release notes	
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